Landing Zone

- **Purpose:** A Landing Zone is a **ready-made and secure AWS environment** that provides a strong foundation for an organization to start using AWS.
  • **Key Points:**
  o Creates a proper **AWS account structure** for different teams, projects, and environments.
  o Keeps important resources like **security, networking, and logging** organized.
  o Provides a **secure and controlled environment** for running applications.
  o Helps maintain the **same security and configuration standards** across AWS accounts.
  o Uses AWS Organizations to **manage multiple AWS accounts centrally**.
  o Helps control who can **access and use AWS resources**.
  o Makes it easier to **add new AWS accounts** as the organization grows.
  o Helps organizations follow **security, compliance, and governance rules**.
  o Acts as the **foundation on which AWS applications and services are built**.

AWS Control Tower

- **Purpose:** AWS Control Tower is a service that **automatically sets up and manages a secure AWS environment with multiple accounts**.
  • **Key Points:**
  o Helps create a **well-organized AWS environment** quickly.
  o Automatically sets up and manages the **Landing Zone**.
  o Uses **guardrails** to make sure AWS accounts follow company security and compliance rules.
  o Makes it easier to **create and manage new AWS accounts**.
  o Provides a **central dashboard** to monitor accounts and their compliance status.
  o Helps detect when an account **does not follow the required rules**.
  o Provides centralized management of **users and access**.
  o Reduces the amount of **manual work** required to manage multiple AWS accounts.
  o Helps organizations maintain **security, consistency, and governance** as their AWS environment grows.

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Easy way to remember

Landing Zone = The Foundation 🏗️
It creates the **proper structure and secure environment** for your AWS organization.

Control Tower = The Manager 👮
It **sets up, monitors, and manages** that environment and makes sure everyone follows the rules.

Simple example:
Imagine a company is building a large office campus:

- **Landing Zone** → Designs the campus with separate buildings for HR, Finance, IT, Security, etc.

- **Control Tower** → Manages the campus, controls access, checks the rules, and makes sure every building follows company policies.

1. EC2 (Elastic Compute Cloud)

- **Purpose**: Scalable virtual servers in the cloud for running applications.
- **Key Points**:
  - Customizable configurations (CPU, memory, storage, networking).
  - Flexible pricing models: On-demand, Reserved, and Spot Instances.
  - Auto Scaling for dynamic resource adjustment based on traffic.
  - Supports Linux, Windows, and custom AMIs.
  - Integrates with VPC, S3, and IAM for better security and networking.

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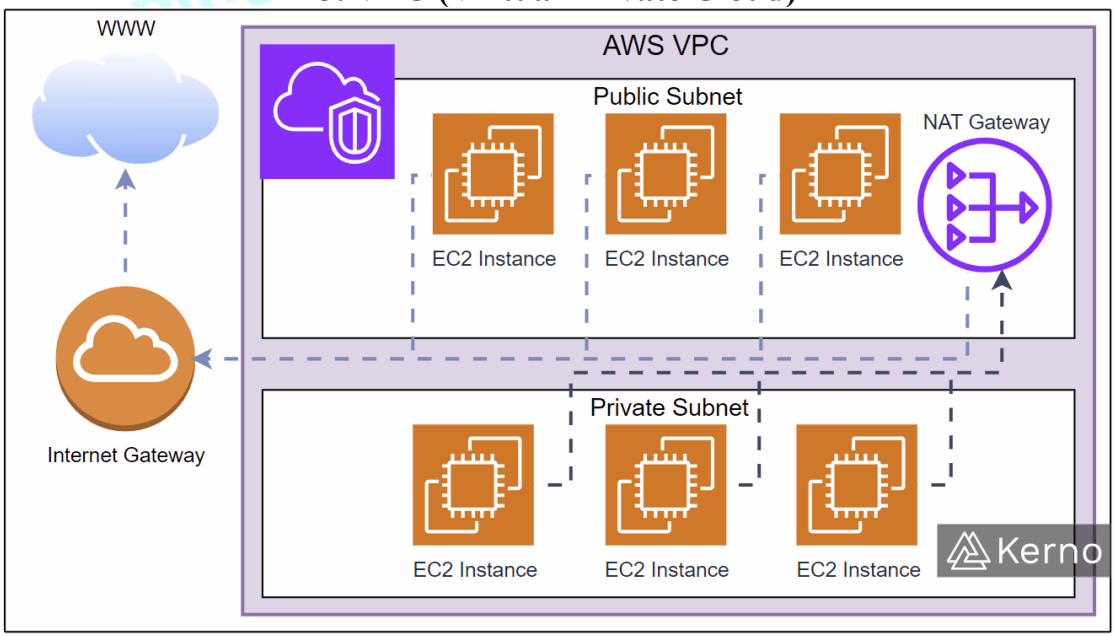
2. IAM (Identity and Access Management)

- **Purpose**: Secure management of access to AWS resources.
- **Key Points**:
  - Creates users, groups, roles, and policies to control permissions.
  - Supports MFA (Multi-Factor Authentication) for enhanced security.
  - Fine-grained access control for different AWS resources.
  - Centrally manages and audits user activities.
  - Works with AWS services to manage permissions across all resources.

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3. VPC (Virtual Private Cloud)



- **Purpose**: Creates isolated networks in AWS for resources.

- **Key Points**:
  - Customizable IP ranges, subnets, and route tables.
  - Provides secure resource communication with security groups and NACLs.
  - Supports VPN and Direct Connect for hybrid cloud setups.
  - Multi-AZ deployment for high availability.
  - Ensures fault tolerance and security for resources.

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4. S3 (Simple Storage Service)

- **Purpose**: Scalable object storage for storing and retrieving data.
- **Key Points**:
  - Virtually unlimited, durable, and cost-effective storage.
  - Multiple storage classes for varying access needs.
  - Built-in encryption for data at rest and in transit.
  - Versioning for object management and restoration.
  - Easily integrates with Lambda, CloudFront, and other AWS services.

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5. Session Manager

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- **Purpose**: Secure shell access to EC2 instances without needing SSH/RDP.
- **Key Points**:
  - Remote management without the need for opening inbound ports.
  - Logging and auditing for session activities.
  - Allows access to EC2 instances in private subnets.
  - Integrates with AWS Systems Manager for automation.
  - No need for public IP addresses on instances.

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6. Route 53

- **Purpose**: DNS service for routing traffic to resources.
- **Key Points**:
  - Domain registration and DNS routing management.
  - Configures routing policies like weighted, latency-based, and geolocation.
  - Health checks and DNS failover to improve availability.
  - Integrates with CloudFront and ELB for dynamic routing.
  - Offers fast DNS resolution for applications.

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7. CloudTrail

- **Purpose**: Monitors and records AWS API calls for auditing.

- **Key Points**:
  - Tracks changes made to AWS resources and user activities.
  - Logs are stored in S3 for security and analysis.
  - Helps with compliance and auditing for regulatory requirements.
  - Integrates with CloudWatch for real-time monitoring.
  - Provides detailed logs for troubleshooting security incidents.

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8. CloudWatch

- **Purpose**: Monitoring and observability service for AWS resources.
- **Key Points**:
  - Collects logs, metrics, and events from AWS resources.
  - Set up alarms to trigger automated actions or notifications.
  - Helps visualize resource performance and system health.
  - Allows for centralized log storage and analysis.
  - Integrates with Auto Scaling and Lambda for automated management.

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9. EKS (Elastic Kubernetes Service)

- **Purpose**: Managed Kubernetes service for containerized applications.
- **Key Points**:
  - Automates the deployment and scaling of containerized applications.
  - Provides integration with AWS services like IAM, VPC, and CloudWatch.
  - Supports multi-cluster and multi-region architectures.
  - Built-in security features like IAM roles for service accounts.
  - Integrates easily with DevOps workflows and CI/CD pipelines.

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10. Auto Scaling

- **Purpose**: Automatically adjusts resources to meet demand.
- **Key Points**:
  - Scales EC2 instances, and other resources based on real-time needs.
  - Ensures cost-efficiency and high availability of applications.
  - Works with Elastic Load Balancer (ELB) to balance traffic.
  - Supports scaling across multiple availability zones.
  - Provides automated scaling for ECS and Lambda functions.

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11. Lambda

- **Purpose**: Serverless compute service that runs code in response to events.
- **Key Points**:

- No need to provision or manage servers.
- Supports multiple languages like Node.js, Python, Java, and Go.
- Pay only for the compute time your code uses.
- Can be triggered by events from S3, DynamoDB, SNS, API Gateway, and more.
- Ideal for event-driven architectures and microservices.

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12. AWS Shield

- **Purpose**: DDoS protection for AWS services.
- **Key Points**:
  - Shield Standard automatically protects AWS customers.
  - Shield Advanced offers enhanced protection, including real-time attack visibility.
  - Protects resources like CloudFront, Route 53, and ELB.
  - Provides 24/7 DDoS response team (DRT) support.
  - Helps mitigate large-scale attacks without affecting performance.

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13. AWS WAF (Web Application Firewall)

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- **Purpose**: Protects web applications from common security threats.
- **Key Points**:
  - Customizable rules to filter malicious web traffic.
  - Protects against SQL injection, cross-site scripting, and bots.
  - Integrates with CloudFront, ALB, and API Gateway.
  - Provides real-time logging of web traffic patterns.
  - Automates security responses using AWS Lambda.

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14. Amazon GuardDuty

- **Purpose**: Threat detection service for AWS environments.
- **Key Points**:
  - Continuously monitors AWS accounts for suspicious activity.
  - Uses machine learning and threat intelligence to detect anomalies.
  - Provides actionable findings to address security risks.
  - Integrates with CloudWatch for automatic response actions.
  - Helps identify unauthorized access, reconnaissance, and data exfiltration.

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15. Amazon Inspector

• **Purpose:** Automated vulnerability management service for AWS workloads.

- **Key Points:**
    - Continuously scans AWS workloads (EC2 instances, container images, and Lambda functions) for security vulnerabilities.
    - Automatically discovers resources and assesses them against known CVEs and security best practices.
    - Provides detailed findings with severity ratings and remediation guidance.
    - Integrates with Amazon EventBridge and CloudWatch for automated alerts and response workflows.
    - Helps identify software vulnerabilities, unintended network exposure, and deviations from security best practices.

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16. AWS Key Management Service (KMS)

- **Purpose**: Manages encryption keys for securing data within AWS.
- **Key Points**:
    - Supports symmetric and asymmetric encryption keys.
    - Easily integrates with AWS services like S3, RDS, and EBS.
    - Automates key rotation to simplify key management.
    - Provides fine-grained access control for encryption keys.
    - Audits key usage via CloudTrail for compliance.

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17. AWS Systems Manager (SSM)

- **Purpose**: Automates operational tasks and manages AWS infrastructure.
- **Key Points**:
    - Manages patching, configurations, and resource compliance.
    - Supports remote management of EC2 instances and on-premises resources.
    - Automates routine administrative tasks like software deployments.
    - Provides centralized management with CloudWatch and CloudTrail.
    - Helps with security and compliance auditing.

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18. Amazon RDS (Relational Database Service)

- **Purpose**: Managed relational database service for multiple database engines.
- **Key Points**:
    - Supports MySQL, PostgreSQL, Oracle, SQL Server, and MariaDB.
    - Automates backups, patch management, and failover.
    - Provides scalability through read replicas and multi-AZ deployments.
    - Offers encryption at rest and in transit for secure data management.
    - Integrates with CloudWatch for monitoring and performance metrics.

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19. Amazon Aurora

- **Purpose**: High-performance relational database compatible with MySQL and PostgreSQL.
- **Key Points**:
  - Provides up to 5x the performance of MySQL and 2x that of PostgreSQL.
  - Self-healing storage and automated backups.
  - Aurora Serverless scales automatically based on demand.
  - Multi-AZ deployments for high availability.
  - Fully managed with automatic patching and scaling.

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20. Amazon DynamoDB

- **Purpose**: Managed NoSQL database service for high-performance applications.
- **Key Points**:
  - Provides fast, low-latency read and write operations.
  - Automatically scales throughput and storage.
  - Supports both key-value and document data models.
  - Built-in encryption at rest for security.
  - Integrates with Lambda, CloudWatch, and other AWS services.

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21. Amazon Redshift

- **Purpose**: Managed data warehouse service for running complex queries.
- **Key Points**:
  - Provides fast SQL-based querying for structured data.
  - Columnar storage and parallel query processing.
  - Scales with Redshift Spectrum for querying data in S3.
  - Integrates with AWS analytics services like Glue and EMR.
  - Offers encryption for data security.

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22. AWS DMS (Database Migration Service)

- **Purpose**: Helps migrate databases to AWS with minimal downtime.
- **Key Points**:
  - Supports migrations between homogeneous and heterogeneous databases.
  - Continuous data replication to ensure data integrity.
  - Monitors migration progress and provides detailed reports.
  - Enables migration of on-premises databases to AWS.

- Supports many database engines, including Oracle, MySQL, and SQL Server.

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23. AWS Batch

- **Purpose**: Efficiently runs batch processing workloads at scale.
- **Key Points**:
  - Automatically provisions the compute resources for batch jobs.
  - Handles thousands of batch computing jobs at any scale.
  - Supports custom compute environments and job scheduling.
  - Integrates with S3 for data storage and processing.
  - Supports EC2 Spot Instances for cost optimization.

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24. AWS Glue

- **Purpose**: Managed ETL service for data preparation and loading.
- **Key Points**:
  - Automatically crawls and discovers datasets.
  - Transforms, cleans, and prepares data for analytics.
  - Integrates with Amazon Redshift, S3, and other data sources.
  - Supports both Python and Scala for scripting.
  - Serverless architecture removes the need for infrastructure management.

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25. Amazon Elastic File System (EFS)

- **Purpose**: Scalable file storage service for EC2 instances and on-premises resources.
- **Key Points**:
  - Provides scalable, shared file storage accessible by EC2 instances.
  - Fully managed with automatic scaling.
  - Supports NFS protocols for integration with on-premises environments.
  - Provides encryption at rest and in transit.
  - Ideal for use cases like content management and big data analytics.

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26. AWS Elastic Beanstalk

- **Purpose**: Platform-as-a-Service (PaaS) for deploying and managing applications.
- **Key Points**:
  - Supports multiple programming languages, including Java, Python, Node.js, and more.
  - Automatically handles deployment, scaling, and monitoring.
  - Simplifies application management with an easy-to-use console.

- Integrates with other AWS services like RDS, S3, and SNS.
- Allows easy customization with configuration options.

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27. Amazon CloudFront

- **Purpose**: Content delivery network (CDN) for faster content delivery.
- **Key Points**:
  - Distributes content to users globally with low latency.
  - Supports dynamic and static content delivery.
  - Can serve from edge locations close to users for faster speeds.
  - Integrates with S3, EC2, and Lambda for dynamic content delivery.
  - Provides DDoS protection and encryption for security.

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28. AWS Snowball

- **Purpose**: Data transport solution for securely transferring large amounts of data to AWS.
- **Key Points**:
  - Provides a physical appliance to transport data to AWS.
  - Useful for large-scale data migrations when internet transfer is impractical.
  - Integrates with S3 and other storage services for data import.
  - Supports encryption for secure data handling.
  - Offers Snowball Edge for local compute and storage capabilities.

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29. AWS CodePipeline

- **Purpose**: Continuous integration and delivery service for automated software workflows.
- **Key Points**:
  - Automates the build, test, and deployment process of applications.
  - Integrates with GitHub, CodeCommit, and other source control repositories.
  - Supports multiple stages for flexible pipeline configurations.
  - Can integrate with Lambda for serverless application deployments.
  - Provides full visibility into your CI/CD process.

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30. AWS CodeBuild

- **Purpose**: Managed build service to compile source code and run tests.
- **Key Points**:
  - Compiles code from source control like GitHub and CodeCommit.
  - Runs tests on your code and produces build artifacts.

- Scales automatically based on build volume.
- Integrates with AWS CodePipeline and other CI/CD tools.
- Supports multiple build environments like Java, Node.js, Python, and Go.

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31. AWS CodeDeploy

- **Purpose**: Automates code deployment to EC2 instances and on-premises servers.
- **Key Points**:
  - Supports rolling, blue/green, and canary deployments.
  - Integrates with AWS services like EC2, Lambda, and ECS.
  - Provides detailed deployment logs for visibility.
  - Enables zero-downtime application deployments.
  - Supports integration with third-party CI/CD tools.

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32. AWS X-Ray

- **Purpose**: Tracing service for debugging and analyzing microservices applications.
- **Key Points**:
  - Provides insights into the performance of distributed applications.
  - Helps identify performance bottlenecks and troubleshoot errors.
  - Supports integration with Lambda, EC2, and other services.
  - Visualizes application architecture and request paths.
  - Helps with security and compliance monitoring.

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7 R	What it means	What you do	Example	Best when
Rehost	Lift & Shift	Move as-is with little/no changes	Move VM to AWS EC2	Need quick migration
Replatform	Lift, Tinker & Shift	Make small improvements	Move DB to Amazon RDS	Want some cloud benefits
Repurchase	Drop & Shop	Replace with a SaaS/product	Replace custom CRM with Salesforce	Existing app is costly to maintain
Refactor	Re-architect	Redesign for cloud-native	Monolith → Microservices	Need maximum cloud benefits
Retire	Remove	Stop using unnecessary applications	Shut down unused reporting system	App has no business value
Retain	Keep	Leave the system where it is	Keep legacy system on-premises	Migration isn't practical yet
Relocate	Move without redesign	Move entire environment/workload	VMware → cloud VMware	Want to move quickly with minimal changes

🧠 Easy memory trick

R R R R R R 😄

- **Rehost** → Move
- **Replatform** → Improve a little
- **Repurchase** → Buy new
- **Refactor** → Redesign
- **Retire** → Remove
- **Retain** → Keep
- **Relocate** → Move the whole environment

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